8. Write a program to implement Linear Regression (LR) algorithm in python.

```
ile  Edit  Format  Run  Options  Window  Help
lataset = [
    [1, 2],
    [2, 4],
    [3, 5],
    [4, 4],
    [5, 5],
    [6, 7],
    [7, 8],
    [8, 9]


: = [row[0] for row in dataset]
' = [row[1] for row in dataset]

. = len(X)

lean_x = sum(X) / n
lean_y = sum(Y) / n

umerator = 0
lenominator = 0

or i in range(n):
    numerator += (X[i] - mean_x) * (Y[i] - mean_y)
    denominator += (X[i] - mean_x) ** 2

l = numerator / denominator

0 = mean_y - b1 * mean_x

rint("Intercept (b0):", b0)
rint("Slope (b1):", b1)

ef predict(x):
    return b0 + b1 * x

rint("\nPredictions:")

or x in X:
    y_pred = predict(x)
    print("Input:", x, "Predicted Output:", round(y_pred,2))

ew_x = 10
ew_y = predict(new_x)

rint("\nNew Input:", new x)
```

```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=========== RESTART: C:/Users/saimo/OneDrive/Desktop/8th program.py ==========
Intercept (b0): 1.4285714285714288
Slope (b1): 0.9047619047619048

Predictions:
Input: 1 Predicted Output: 2.33
Input: 2 Predicted Output: 3.24
Input: 3 Predicted Output: 4.14
Input: 4 Predicted Output: 5.05
Input: 5 Predicted Output: 5.95
Input: 6 Predicted Output: 6.86
Input: 7 Predicted Output: 7.76
Input: 8 Predicted Output: 8.67

New Input: 10
Predicted Value: 10.48
>>>
=================== RESTART: C:/Users/saimo/OneDrive/Desktop/8th program.py ===================
Intercept (b0): 1.4285714285714288
Slope (b1): 0.9047619047619048

Predictions:
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Input: 7 Predicted Output: 7.76
Input: 8 Predicted Output: 8.67

New Input: 10
Predicted Value: 10.48
>>>
Ln: 37  Col: 0
```

Ln: 43 Col: 0